Nutrition

Concepts and Ideas

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Learning Objectives:

- Outline the main types of nutrients.
- Describe how to select foods to maintain a healthy diet.
- Classify the vitamins according to their solubility.
- Explain the importance of balanced vitamin intake.

- Nutrients are chemical substances in food that body cells use for growth, maintenance, and repair.

- The six main types of nutrients are water, carbohydrates, lipids, proteins, minerals, and vitamins.

- **Water** is the nutrient needed in the largest amount—about 2–3 liters per day.

- As the most abundant compound in the body, water provides the medium in which most metabolic reactions occur, and it also participates in some reactions (for example, hydrolysis reactions).

- *Essential nutrients* are specific nutrient molecules that the body cannot make in sufficient quantity to meet its needs and thus must be obtained from the diet.

- Some amino acids, fatty acids, vitamins, and minerals are essential nutrients.

Carbohydrates

- Carbohydrates include sugars, glycogen, starches, and cellulose.
- Even though they are a large and diverse group of organic compounds and have several functions, carbohydrates represent only 2–3% of your total body mass.
- The three major groups of carbohydrates, based on their sizes, are monosaccharides, disaccharides, and polysaccharides.
- Monosaccharides and Disaccharides are known as simple sugars.
- A disaccharide is a molecule formed from the combination of two monosaccharides.

Carbohydrates

- The main polysaccharide in the human body is glycogen.
- Starches are polysaccharides formed from glucose by plants. They are found in foods such as pasta and potatoes and are the major carbohydrates in the diet.
- Cellulose is a polysaccharide formed from glucose by plants that cannot be digested by humans but does provide bulk to help eliminate feces.

Lipids

- A second important group of organic compounds is lipids.

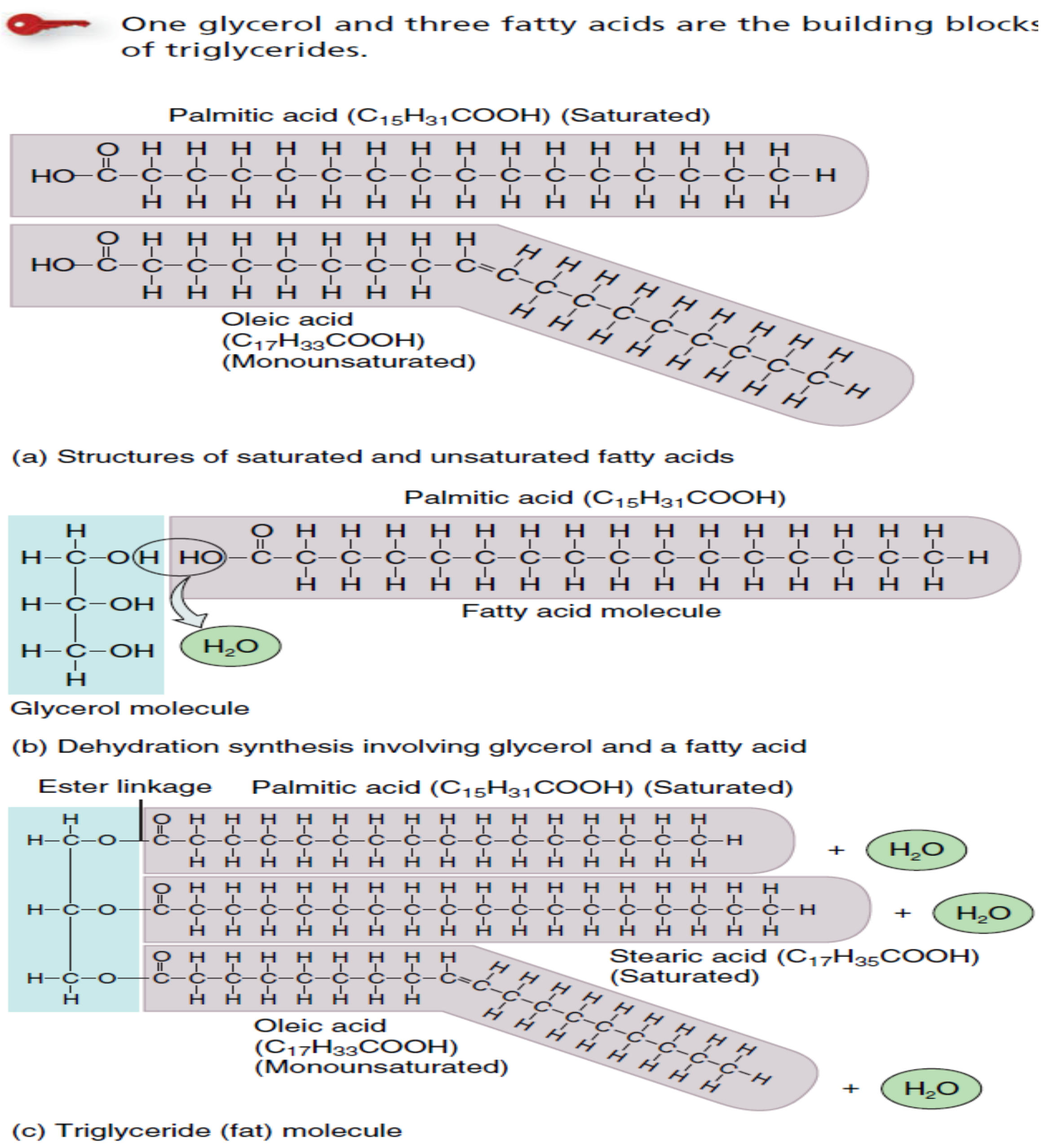
- The diverse lipid family includes: fatty acids, triglycerides (fats and oils), phospholipids, steroids, eicosanoids, and a variety of other substances, including fat-soluble vitamins (vitamins A, D, E, and K) and lipoproteins.

- Among the simplest lipids are the fatty acids, which are used to synthesize triglycerides and phospholipids.

Fatty Acids

- A saturated fatty acid contains only single covalent bonds between the carbon atoms of the hydrocarbon chain. Because they lack double bonds, each carbon atom of the hydrocarbon chain is saturated with hydrogen atoms .

- An unsaturated fatty acid contains one or more double covalent bonds between the carbon atoms of the hydrocarbon chain. Thus, the fatty acid is not completely saturated with hydrogen atoms.



- The unsaturated fatty acid has a kink (bend) at the site of the double bond.

- If the fatty acid has just one double bond in the hydrocarbon chain, it is monounsaturated and it has just one kink.

- If a fatty acid has more than one double bond in the hydrocarbon chain, it is polyunsaturated and it contains more than one kink.

Triglycerides

- The most plentiful lipids in your body and in your diet are the triglycerides.
- A triglyceride consists of a single glycerol molecule and three fatty acid molecules.

Fat

- A fat is a triglyceride that is a solid at room temperature.
- The fatty acids of a fat are mostly saturated. Because these saturated fatty acids lack double bonds in their hydrocarbon chains, they can closely pack together and solidify at room temperature.
- Although saturated fats occur mostly in meats (especially red meats) and nonskim dairy products (whole milk, cheese, and butter), they are also found in a few plant products, such as palm oil, and coconut oil.
- Diets that contain large amounts of saturated fats are associated with disorders such as heart disease and colorectal cancer.

oil

- An oil is a triglyceride that is a liquid at room temperature.
- The fatty acids of an oil are mostly unsaturated.
- The kinks at the sites of the double bonds prevent the unsaturated fatty acids of an oil from closely packing together and solidifying.
- The fatty acids of an oil can be either monounsaturated or polyunsaturated.

oil

- Olive oil, peanut oil, canola oil, most nuts, and avocados are rich in triglycerides with monounsaturated fatty acids.

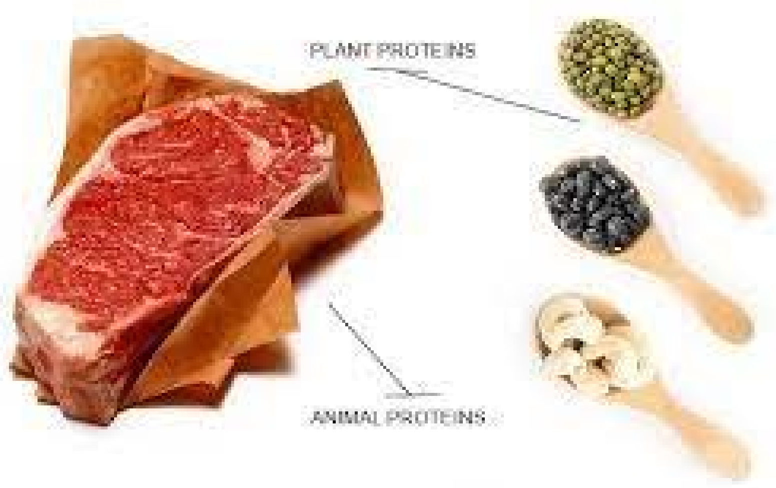
- Corn oil, sunflower oil, soybean oil, and fatty fish (salmon, tuna) contain a high percentage of polyunsaturated fatty acids.

- Both monounsaturated and polyunsaturated fats are believed to decrease the risk of heart disease.

Proteins

- Much more complex in structure than carbohydrates or lipids.

- Proteins have many roles in the body and are largely responsible for the structure of body tissues.



What is a Calorie?

- A calorie (cal) is the amount of heat required to raise the temperature of 1 gram of water 1C.

- A kilocalorie equals 1000 calories.

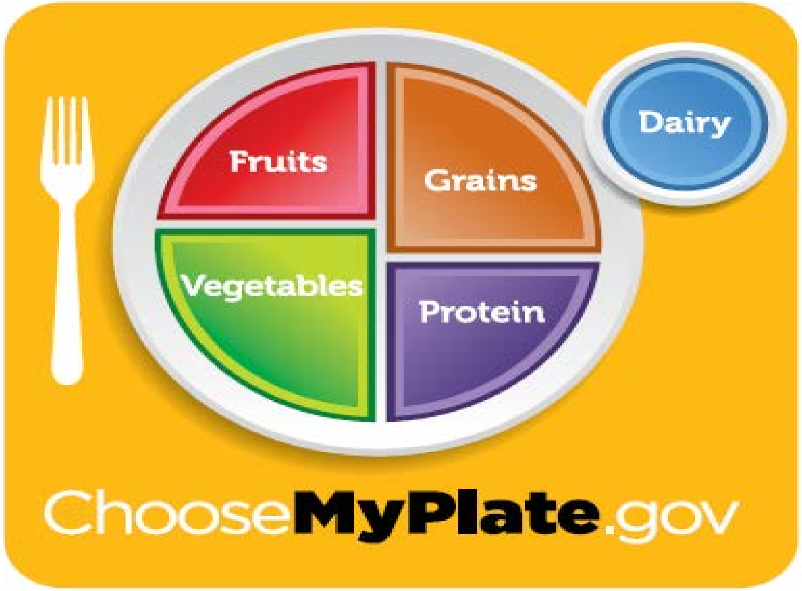
- Each gram of protein or carbohydrate in food provides about 4 Calories; 1 gram of fat (lipids) provides about 9 Calories.

- BMR is 1200–1800 Cal/day in adults, or about 24 Cal/kg of body mass in adult males and 22 Cal/kg in adult females.

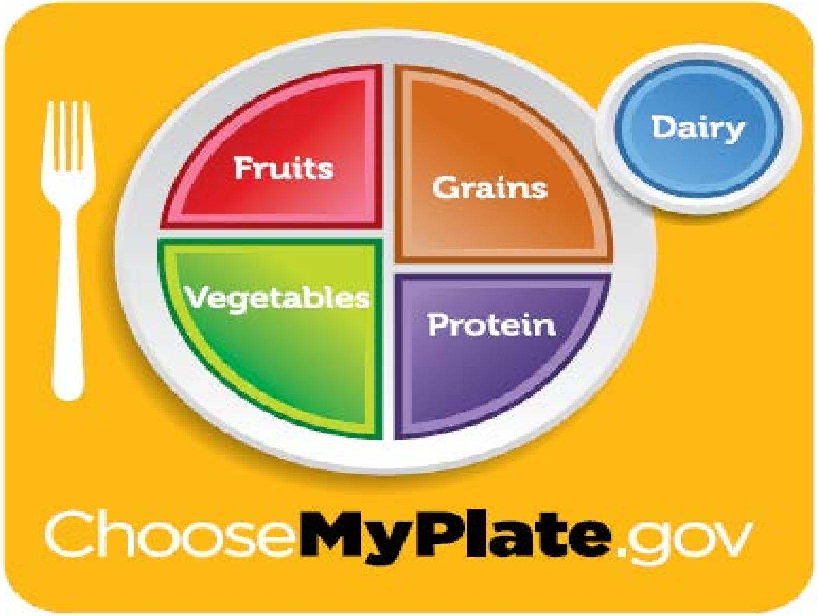
- The added calories needed to support daily activities, such as digestion and walking, range from 500 Cal for a small, relatively sedentary person to over 3000 Cal for a person in training for Olympic-level competitions.

How to select foods to maintain a healthy diet?

- The recommend distribution of calories:
- 50–60% from carbohydrates, with less than 15% from simple sugars; less than 30% from fats (triglycerides are the main type of dietary fat), with no more than 10% as saturated fats; and about 12–15% from proteins.



- Healthy eating emphasis on proportionality, variety, moderation, and nutrient density in a healthy diet.
- Proportionality simply means eating more of some types of foods than others.
- Variety is important for a healthy diet because no one food or food group provides all of the nutrients and food types that the body needs.
- No single food contains all of the required vitamins—one of the best reasons to eat a varied diet.



Healthy Tips

- include making half of your grains whole grains
- choosing whole or cut-up fruits more often than juice
- selecting fat-free or low-fat dairy products
- keeping meat and poultry portions small and lean

Minerals:

- Are inorganic elements that occur naturally in the earth's crust.
- In the body they appear in combination with one another, in combination with organic compounds, or as ions in solution.

Minerals:

- The body generally uses the ions of the minerals rather than the non-ionized form. Some minerals, such as chlorine, are toxic or even fatal if ingested in the non-ionized form.
- Typical diets supply adequate amounts of potassium, sodium, chloride, and magnesium.
- Some attention must be paid to eating foods that provide enough calcium, phosphorus, iron, and iodide. Excess amounts of most minerals are excreted in the urine and feces.

Vitamins:

- Vitamins are Organic nutrients required in small amounts to maintain growth and normal metabolism, they do not provide energy or serve as the body's building materials.

- Most vitamins with known functions are coenzyme.

- Most vitamins cannot be synthesized by the body and must be ingested in food.

- Other vitamins, such as vitamin K, are produced by bacteria in the GI tract and then absorbed.

- The body can assemble some vitamins if the raw materials, called provitamins, are provided e.g: vitamin A is produced by the body from the provitamin beta-carotene, a chemical present in yellow vegetables such as carrots.

Classifications of Vitamins:

Vitamins are divided into two main groups:

1. The fat-soluble vitamins: include vitamin A, D, E, and K.

- Are absorbed along with other dietary lipids in the small intestine and packaged into chylomicrons.
- They cannot be absorbed in adequate quantity unless they are ingested with other lipids.

Classifications of Vitamins:

2. The water-soluble vitamins: include several B vitamins and vitamin C.

- Are dissolved in body fluids.
- Excess quantities of these vitamins are not stored but instead are excreted in the urine

Vitamins as Antioxidants:

- Antioxidant is a substance that inactivates oxygen derived free radicals.
- In addition to their other functions, three vitamins—C, E, and beta-carotene (a provitamin) are termed antioxidant vitamins because they inactivate oxygen free radicals.

- Free radicals are highly reactive ions or molecules that can damage cell membranes, DNA, and other cellular structures and contribute to the formation of artery-narrowing atherosclerotic plaques.

- Some free radicals arise naturally in the body, and others come from environmental hazards such as tobacco smoke and radiation.

- Antioxidant vitamins are thought to play a role in protecting against some kinds of cancer, reducing the buildup of atherosclerotic plaque, delaying some effects of aging.

Clinical Connection..

- It is recommended to eat a balanced diet that includes a variety of foods rather than taking vitamin or mineral supplements, except in special circumstances.

- Common examples of necessary supplementations include

- iron for women who have excessive menstrual bleeding; iron and calcium for women who are pregnant or breast-feeding;
- folic acid (folate) for all women who may become pregnant, to reduce the risk of fetal neural tube defects;
- calcium for most adults, because they do not receive the recommended amount in their diets;
- vitamin B12 for strict vegetarians, who eat no meat.

Questions and Comments?